

Performance Testing

Date	15 March 2026
Team ID	
Project Name	OptiCrop
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust / manual timing
Type of Testing	Load Testing
Target Module / API	/predict endpoint (crop prediction)
Test Environment	Cloud (Vercel serverless)
Test Date	05 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single prediction request	1	10	Crop returned < 2s
2	Concurrent farmers	50	60	All requests succeed
3	Repeated rapid requests	100	60	Stable, no errors
4	Invalid input handling	10	30	Graceful error message

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	~150 ms	Pass	Well under target
2	Response Time (Max)	< 5 seconds	~380 ms	Pass	Peak load acceptable
3	Throughput (Req/sec)		~65 req/s	Pass	Serverless scales
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	~40%	Pass	Low CPU use
6	Memory Utilization	< 80%	~55%	Pass	Within limit

Step 4: Observations & Analysis

Key Findings:
 The predict endpoint averaged ~150 ms with 0% error rate. Model accuracy is 97.27%. The serverless deployment scaled smoothly under concurrent load; CPU and memory stayed well below thresholds.
 [Describe what the test results revealed about your system's performance]

Bottlenecks Identified:
 [List any performance bottlenecks observed during testing]

Optimization Steps Taken:
 [Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here]

[Insert Screenshot 2 here]